NUMERICAL ANALYSIS
PROBLEM SET 4

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1. Problem 1

1.a. We are deriving the 2-point open Newton-Coats formula

$$\int_a^b f(x)dx \approx I_{2,\text{open}}(f) = \frac{b-a}{3}\Big(2f(a+h) - f(a+2h) + 2f(a+3h)\Big)$$

for evenly spaced points with $h = \frac{b-a}{n+2}$. Now we set $n = 2$ so that $h = \frac{b-a}{4}$, and we need to sort out

$$I_{2,\text{open}}(f) = \int_a^b \sum_{i=0}^{2} L_{2,i}(x) f(x_i) dx. \tag{1}$$

But we need to shift to an easier or simpler variable to work with. So we set $x = a + t \cdot h$, with $dx = h \cdot dt$. Now as x moves from a to b we see that $(b = a + 4h)$, t moves from 0 to 4, and we have

$$\int_a^b f(x)dx = h \int_0^4 f(a+ht)dt,$$

which we interpolate $f(a+ht)$ at $t = 1, 2, 3$. We write out the Lagrangian polynomials $L_{2,0}, L_{2,1}, L_{2,2}$ for the sake of the problem. This follows as

$$L_{2,0}(x) = \frac{(t-2)(t-3)}{(1-2)(1-3)} = \frac{(t-2)(t-3)}{2} = \frac{1}{2}(t^2 - 5t + 6)$$

$$L_{2,1}(x) = \frac{(t-1)(t-3)}{(2-1)(2-3)} = -(t-1)(t-3) = -t^2 + 4t - 3$$

$$L_{2,2}(x) = \frac{(t-1)(t-2)}{(3-1)(3-2)} = \frac{(t-1)(t-2)}{2} = \frac{1}{2}(t^2 - 3t + 2).$$

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And we then need to integrate these Lagrange interpolants, which we do as

$$\int_0^4 \frac{1}{2}(t^2 - 5t + 6)dt = \frac{t^3}{6} - \frac{5t^2}{4} + 3t\bigg|_0^4 = \frac{8}{3}$$

$$\int_0^4 -t^2 + 4t - 3dt = -\frac{t^3}{3} + 2t^2 - 3t\bigg|_0^4 = -\frac{4}{3}$$

$$\int_0^4 \frac{1}{2}(t^2 - 3t + 2)dt = \frac{t^3}{6} - \frac{3t^2}{2} + 2t\bigg|_0^4 = \frac{8}{3}$$

Thus, when we plug this into equation (1), we get the following

$$\begin{aligned}
\int_a^b f(x)dx &\approx h\left(\frac{8}{3}f(a+h) - \frac{4}{3}f(a+2h) + \frac{8}{3}f(a+3h)\right) \\
&\approx \frac{4h}{3}\left(2f(a+h) - f(a+2h) + 2f(a+3h)\right) \\
&\approx \frac{b-a}{3}\left(2f(a+h) - f(a+2h) + 2f(a+3h)\right).
\end{aligned}$$

which solves our problem

1.b. We must show that our quadrature is exact for $1, x, x^2, x^3$ but not for x^4. Again we rescale the interval in teh fashion

$$x = a + ht, \qquad h = \frac{b-a}{4}, x \in [a,b] \to t \in [0,4]$$

So $Q(g) = \frac{4}{3}\left(2g(1) - g(2) - 2g(3)\right)$. But, we need to check if $I(g) = Q(g)$ for $g(t) \in \{1, x, x^2, x^3, x^4\}$. So we write

$$\int_0^4 1dt = 4 \stackrel{?}{=} \frac{4}{3}(2 - 1 + 2) = 4 \quad \checkmark$$

$$\int_0^4 tdt = \frac{t^2}{2}\bigg|_0^4 = 8 \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 2 + 2\cdot 3) = 8 \quad \checkmark$$

$$\int_0^4 t^2dt = \frac{t^3}{3}\bigg|_0^4 = \frac{64}{3} \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 4 + 2\cdot 9) = \frac{4\cdot 16}{3} = \frac{64}{3} \quad \checkmark$$

$$\int_0^4 t^3dt = \frac{t^4}{4}\bigg|_0^4 = 64 \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 8 + 2\cdot 27) = \frac{4\cdot 48}{3} = \frac{192}{3} = 64 \quad \checkmark$$

$$\int_0^4 t^4dt = \frac{t^5}{5}\bigg|_0^4 = \frac{1024}{5} = 204.8 \stackrel{?}{=} \frac{4}{3}(2 - 32 + 2\cdot 243) = \frac{1824}{3} = 608 \quad \times$$

So we are done.

2. Problem 2

Let

$$I(f) = \int_a^b f(x)dx$$

and let $[a, b]$ be divided into subintervals in the following fashion

$$\{[a, x_1 + h], [x_2 - h, x_2 + h], \ldots, [x_n - h, b]\},$$

with $x_1, \ldots, x_n$ being the midpoints of the subintervals. We thusly see that $h = \frac{b-a}{2n}$, that $x_j = a + (2j - 1)h$, and that each of our subintervals has width $2h$. Now consider the j^{th} subinterval $[x_j - h, x_j + h]$. We can write $f(x)$ over the interval using Taylor's theorem as

$$f(x) = f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2,$$

for some η_x between x and x_j. So we integrate

$$\begin{aligned}
\int_{x_j-h}^{x_j+h} f(x)dx &= \int_{x_j-h}^{x_j+h} \left(f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2 \right) dx \\
&= \int_{x_j-h}^{x_j+h} f(x_j)dx + \int_{x_j-h}^{x_j+h} f'(x_j)(x - x_j)dx + \int_{x_j-h}^{x_j+h} \frac{f''(\eta_x)}{2}(x - x_j)^2 dx \\
&= 2hf(x_j) + 0 + \frac{f''(\xi_j)}{2} \int_{x_j-h}^{x_j+h} (x - x_j)^2 dx,
\end{aligned}$$

for some ξ_j clearly dependent upon our choice of η_x. But notice that our last integral can be manipulated by a transformation of variables in the following fashion

$$\int_{x_j-h}^{x_j+h} (x - x_j)^2 dx = \int_{-h}^{h} t^2 dt = \frac{2h^3}{3}.$$

Substituting that back in to the over all equation we see that

$$\int_{x_j-h}^{x_j+h} f(x)dx + \frac{f''(\xi_j)}{\cancel{2}} \frac{\cancel{2}h^3}{3}.$$

So we take this result and sum across all the subintervals to get that

$$\int_b^a f(x)dx = \sum_{j=1}^{n} \int_{x_j-h}^{x_j+h} f(x)dx = 2h \sum_{j=1}^{n} f(x_j) + \frac{h^3}{3} \sum_{j=1}^{n} f''(\xi_j).$$

But we notice that $\sum_{j=1}^{n} f''(\xi_j) = nf''(\xi)$ with $f''(\xi)$ representing the average of all $f''(\xi_j)$. And with some careful substitution, along with noticing that $n = \frac{b-a}{2h}$ by manipulating our definition of h, we see that

$$\int_a^b f(x)dx = 2h \sum_{j=1}^{n} f(x_j) + \frac{(b-a)h^2}{6} f''(\xi),$$

where $\xi \in [a, b]$, and we are done.

3. Problem 3

We will determine the weights w_i and the quadriture points x_i in the three point Guass-Legendre quadrature formula

$$\int_{-1}^{1} f(x)dx \approx w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3).$$

Now by definition the degree of precision of this formula just falls out by us having to generate $2n-1$ equations, so that we can use the method of undetermined coefficients for finding the weights. This yields a $2(3)-1=5$ degree of precision. So we go through the process of generating the equations over which we'll solve for the weights by using the integrals of $1, x, x^2, x^3, x^4, x^5$, as

$$f(x) = 1: \quad w_1 + w_2 + w_3 = \int_{-1}^{1} 1 \cdot dx = 2,$$

$$f(x) = x: \quad w_1 x + w_2 x + w_3 x = \int_{-1}^{1} x dx = \left.\frac{x^2}{2}\right|_{-1}^{1} = 0,$$

$$f(x) = x^2: \quad w_1 x^2 + w_2 x^2 + w_3 x^2 = \int_{-1}^{1} x^2 dx = \left.\frac{x^3}{3}\right|_{-1}^{1} = \frac{2}{3},$$

$$f(x) = x^3: \quad w_1 x^3 + w_2 x^3 + w_3 x^3 = \int_{-1}^{1} x^3 dx = \left.\frac{x^4}{4}\right|_{-1}^{2} = 0,$$

$$f(x) = x^4: \quad w_1 x^4 + w_2 x^4 + w_3 x^4 = \int_{-1}^{1} x^4 dx = \left.\frac{x^5}{5}\right|_{-1}^{1} = \frac{2}{5}$$

$$f(x) = x^5: \quad w_1 x^5 + w_2 x^5 + w_3 x^5 = \int_{-1}^{1} x^5 dx = \left.\frac{x^6}{6}\right|_{-1}^{1} = 0.$$

Our goal now is to define a polynomial, call it $p(x)$, that vanishes at the quadriture points x_1, x_2, x_3. So we have

$$\begin{aligned} p(x) &= (x - x_1)(x - x_2)(x - x_3) \\ &= x^3 - x_1 x^2 - x_2 x^2 - x_3 x^2 + x_1 x_2 x + x_1 x_3 x + x_2 x_3 x - x_1 x_2 x_3 \\ &= x^3 s_1 x^2 + s_2 x + s_3, \end{aligned}$$

where $s_1 = -(x_1 + x_2 + x_3)$, $s_2 = x_1 x_2 + x_1 x_3 + x_2 x_3$, and $s_3 = -x_1 x_2 x_3$.

Now to find our weights we set $p(x) = 0$, then $xp(x) = 0$, and finally $x^2p(x) = 0$. So let us begin with

$$\begin{aligned}
p(x_i) = 0 &= x_i^3 + s_1 x_i^2 + s2x_i + s_3 = 0 \\
&= \sum w_i x_i^3 + s_1 \sum w_i x_i^2 + s_2 \sum w_i x_i + s3 \sum w_i = 0 \\
&= 0 + s_1 \frac{2}{3} + 0 + s_3 \cdot (-2)
\end{aligned}$$

So we have that

$$s_1 = -3s_3.$$

Now set $xp(x) = 0$, and let us see what we can find:

$$\begin{aligned}
xp(x) &= x^4 + s_1 x^3 + s_2 x^2 + s_3 x = 0 \\
&= \sum w_i x_i^4 + s_1 \sum w_i x_i^3 + s_2 \sum w_i x_i^2 + s_3 \sum w_i x_i = 0 \\
&= \frac{2}{5} + s_1 \cdot 0 + s_2 \frac{2}{3} + s_3 \cdot 0 = 0 \\
&= \frac{2}{5} + s_2 \frac{2}{3} = 0 \quad \Rightarrow \quad s_2 = -\frac{3}{5}.
\end{aligned}$$

Next we set $x^2p(x) = 0$ to again have another equation over which we can solve for s_1, s_2, s_3, which follows as

$$\begin{aligned}
x^2p(x) &= x^5 + s_1 x^4 + s_2 x^3 + s_3 x^2 = 0 \\
&= \sum w_i x_i^5 + s_1 \sum w_i x_i^4 + s_2 \sum w_i x_i^3 + s_3 \sum w_i x_i^2 = 0 \\
&= 0 + s_1 \frac{2}{5} + s_2 \cdot 0 + s_3 \frac{2}{3} = 0 \\
&= s_1 \frac{2}{5} + s_3 \frac{2}{3}.
\end{aligned}$$

Because we have $s1 = -3s_3$ we see that $-6s_3/5 + 2s_3/3 = s_3(2/3 - 6/5) = 0$. So, $s_3 = 0$ and thus $s_1 = 0$, which setting $p(x) = 0$ yields

$$p(x) = x^3 - \frac{3}{5}x = x\left(x^2 - \frac{3}{5}\right) = 0.$$

So we see that

$$x_1 = -\sqrt{\frac{3}{5}}, \qquad x_2 = 0, \qquad x_3 = \sqrt{\frac{3}{5}}.$$

So now we solve for the weights using our first three $f(x)$ equations from the beginning. We begin with,

$$\sum w_i = w_1 + w_2 + w_3 = 2. \tag{2}$$

Next, we see that

$$\sum w_i x_i = -w_1\sqrt{\frac{3}{5}} + w_2 \cdot 0 + w_3\sqrt{\frac{3}{5}} = 0 \qquad \Longleftrightarrow \qquad w_1 = w_3.$$

Lastly we compute,

$$\sum w_i x_i^2 = w_1\frac{3}{5} + w_3\frac{3}{5} = \frac{6w_1}{5} = \frac{2}{3} \qquad \Longleftrightarrow \qquad w_1 = w_3 = \frac{5}{9}$$

Which plugging back into equation (2) yields

$$\frac{5}{9} + w_2 + \frac{5}{9} = 2 \qquad \Longleftrightarrow \qquad w_2 = \frac{8}{9}.$$

So we see that

$$\int_{-1}^{1} f(x)dx \approx \frac{5}{9}f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9}f(0) + \frac{5}{9}f\left(\sqrt{\frac{3}{5}}\right),$$

and we are done.

4. Problem 4

Consider the initial value problem given as

$$y' = y\cos(t), \qquad y(0) = 1, \qquad t \in [0, 2\pi].$$

4.a. We need to show existence and uniqueness of the solution of some subinterval $I \subseteq [0, 2\pi]$ containing $t = 0$. To do this we show the Lipschitz condition $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$. First we know that $f(x, y)$ must be continuous in our domain because it is the product of two continuous functions. Next we notice that

$$|f(t, y_1) - f(t, y_2)| = |y_1\cos(t) - y_2\cos(t)| = |\cos(t)||y_1 - y_2| \leq 1 \cdot |y_1 - y_2|,$$

because $|\cos(t)| \leq 1$ for all $t \in \mathbb{R}$. So our function is Lipschitz with $L = 1$. Furthermore, we have a theorem that states that if we are Lipschitz on our domain then for an initial value problem of our form, then we are guaranteed existence and uniqueness of the solution over our interval.

4.b. We now compute the Forward Euler Method for solving the problem. Specifically, we need to work out the stepwise solution for $y_{n+1} = y_n + hf(t, y_n)$. Set $h = \pi/4$ with $t_0 = 0$, $y_0 = 1$, $t_1 = \pi/4$, and $t_2 = \pi/2$.

Step 1. With $t_0 = 0$, $t_1 = \pi/4$, and $f(t_0, y_0) = 1 \cdot \cos(0) = 1$, we see that

$$y_1 = y_0 + hf(t_0, y_0) = 1 + \frac{\pi}{4}(1) = 1 + \frac{\pi}{4}$$

Step 2. Now that we have y_1 above, we use $t_1 = \pi/4$ and $t_2 = \pi/2$. We then notice that

$$f(t_1, y_1) = \left(1 + \frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}\right) = \left(1 + \frac{\pi}{4}\right)\frac{\sqrt{2}}{2}.$$

Then we compute y_2 in the following manner

$$y_2 = y_1 + hf(t_1, y_1) = \left(1 + \frac{\pi}{4}\right) + \left(1 + \frac{\pi}{4}\right)\frac{\sqrt{2}}{2} \cdot \frac{\pi}{4} = \left(1 + \frac{\pi}{4}\right)\left(1 + \frac{\pi\sqrt{2}}{8}\right) \approx 2.7554.$$

4.c. Directly solving the initial value problem for $y' = y\cos(t)$ follows as

$$\begin{aligned} \frac{dy}{y} &= \cos(t) \\ \int \frac{dy}{y} &= \int \cos(t)dt \\ \ln(y) &= \sin(t) + c \\ y &= e^{\sin(t)+c} = ce^{\sin(t)}. \end{aligned}$$

Then we substitute in $y(0) = 1$ as

$$y(0) = ce^{\sin(0)} = 1 \Leftrightarrow c = 1.$$

Thus, we have that

$$y(t) = e^{\sin(t)}.$$

We were asked to show that $y(t)$ is bounded in the fashion $e^{-\pi} \leq y \leq e^{\pi}$, but $-1 \leq \sin(t) \leq 1$ so naturally $e^{-\pi} \leq e^{-1} \leq y(t) \leq e^{1} \leq e^{\pi}$. The rest of (c) we omit for the sake of expedience.

4.d. The backwards Euler formula follows as $y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$. So we write and solve for y_{n+1}.

$$\begin{aligned} y_{n+1} &= y_n + hy_{n+1}\cos(t_{n+1}) \\ y_{n+1}(1 - h\cos(t_{n+1})) &= y_n \\ y_{n+1} &= \frac{y_n}{1 - h\cos(t_{n+1})} \end{aligned}$$

Furthermore, we are given $h = \pi/4$ with $t_0 = 0$, $y_0 = 1$, $t_1 = \pi/4$, and $t_2 = \pi/2$. So we do our two steps.

Step 1.

$$y_1 = \frac{y_0}{1 - h\cos(t_1)} = \frac{1}{1 - \frac{\pi}{4}\cos\left(\frac{\pi}{4}\right)} = \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}.$$

Step 2.

$$y_2 = \frac{y_1}{1 - h\cos(t_2)} = \frac{\left(\frac{1}{1-\frac{\pi\sqrt{2}}{8}}\right)}{1 - \frac{\pi}{4}\cos\left(\frac{\pi}{2}\right)} = y_1.$$

So

$$y\left(\frac{\pi}{2}\right) \approx \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}$$

4.e. We ran plots of the Euler's methods and such. The code can be found in:
https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet4

The exact solution for the problem can be found back in part (c) of the problem. That said, we have generated a plot of the forward and backwards Euler's method given in Figure 1

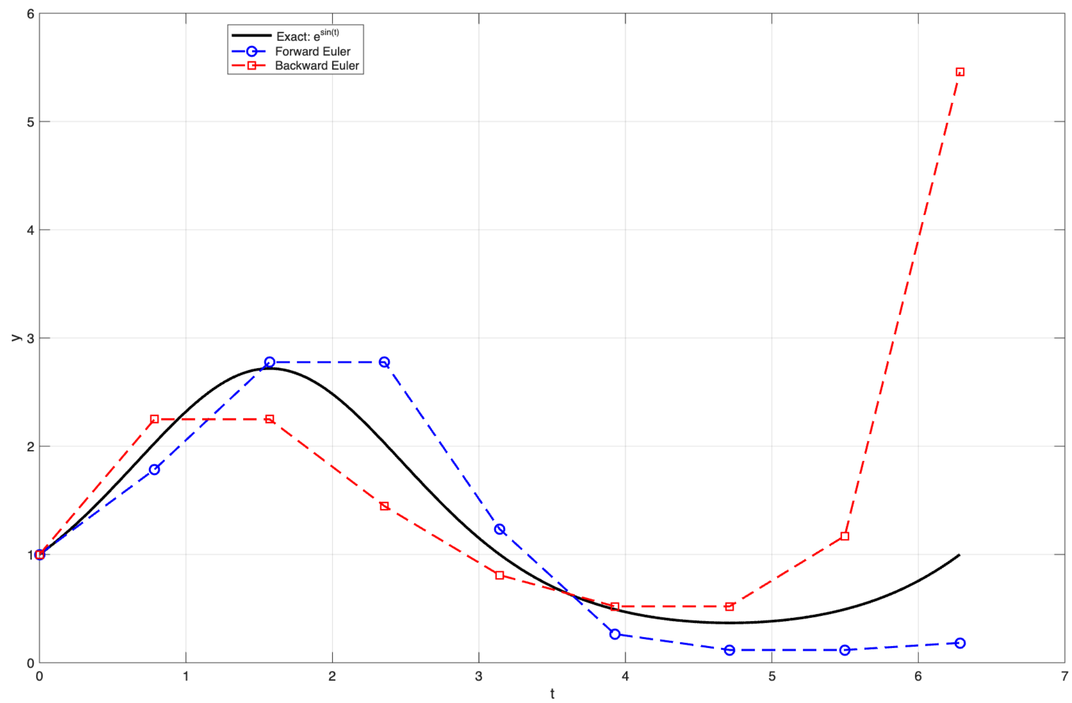

FIGURE 1. For $y' = y\cos(t)$, a plot of the exact solution along with Euler's backwards and forwards methods.

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We've omitted talking about the Global Error Theorem for Euler's method out of the need for expedience.

5. Problem 5

Consider the initial value problem given as

$$y' = \sin(t) + y, \qquad y(0) = 0, \qquad t \in [0, 1].$$

5.a. We go about finding the exact solution of the initial value problem as

$$dy = \sin(t)dt + y$$
$$dy - y = \sin(t)dt$$

Now we are going to use an integrating factor $\mu(t) = e^{-t}$ so we multiply both sides by $\mu(t)$ which becomes

$$e^{-t}dy - ye^{-t} = e^{-t}\sin(t)dt.$$

Notice that the left hand side is the result of a product rule differentiation. Now we also know that,

$$\frac{d}{dt}\left[e^{-t}y\right] = e^{-t}dy - ye^{-t}.$$

Which allows us to do the following

$$\begin{aligned} e^{-t}y &= \int e^{-t}\sin(t)dt \\ e^{-t}y &= -\frac{1}{2}e^{-t}\Bigg(\sin(t)+\cos(t)\Bigg)+C \\ y(t) &= -\frac{1}{2}(\sin(t)+\cos(t))+C. \end{aligned}$$

Now we solve for C by using our initial condition $y(0) = 0$ as:

$$y(0) = -\frac{1}{2}(\sin(0)+\cos(0)) + C = 0 \qquad \Longleftrightarrow \qquad C = \frac{1}{2}$$

So we have that

$$y(t) = -\frac{1}{2}(\sin(t)+\cos(t)) + \frac{1}{2}.$$

We not want to find $y(1)$, which follows as:

$$y(1) = -\frac{1}{2}\sin(1) - \frac{1}{2}\cos(1) \approx -0.1908866453\ldots$$

5.b. For the next portion of this problem we ran the following code

```
% Forward Euler for y' = sin(t) + y, y(0) = 0 on [0,1]
clear; clc;
f = @(t,y) sin(t) + y;
yexact = @(t) 0.5*(exp(t) - sin(t) - cos(t));

a = 0;
b = 1;
Nh = 2;
h = (b-a)/Nh;

t = zeros(Nh+1,1);
y = zeros(Nh+1,1);

% initial condition
t(1) = 0;
y(1) = 0;

% Forward Euler method
for n = 1:Nh
    t(n+1) = t(n) + h;
    y(n+1) = y(n) + h*f(t(n),y(n));
end

% exact value and error at t=1
y_exact_at_1 = yexact(1);
```

```
error_at_1 = abs(y(end) - y_exact_at_1);

fprintf('Forward Euler approximation at t=1: %.10f\n', y(end));
fprintf('Exact value at t=1: %.10f\n', y_exact_at_1);
fprintf('Absolute error at t=1: %.10f\n', error_at_1);
```

And we achieved the following output:

```
Forward Euler approximation at t=1: 0.2397127693
Exact value at t=1: 0.6682542689
Absolute error at t=1: 0.4285414996
```

5.c. We then went about doing forwards and backwards Euler's method on the problem to see how it did, and we achieved the following graphs. For forward Euler's method we have Figure 2 which has a constant slope to it so we would expect linear convergence.

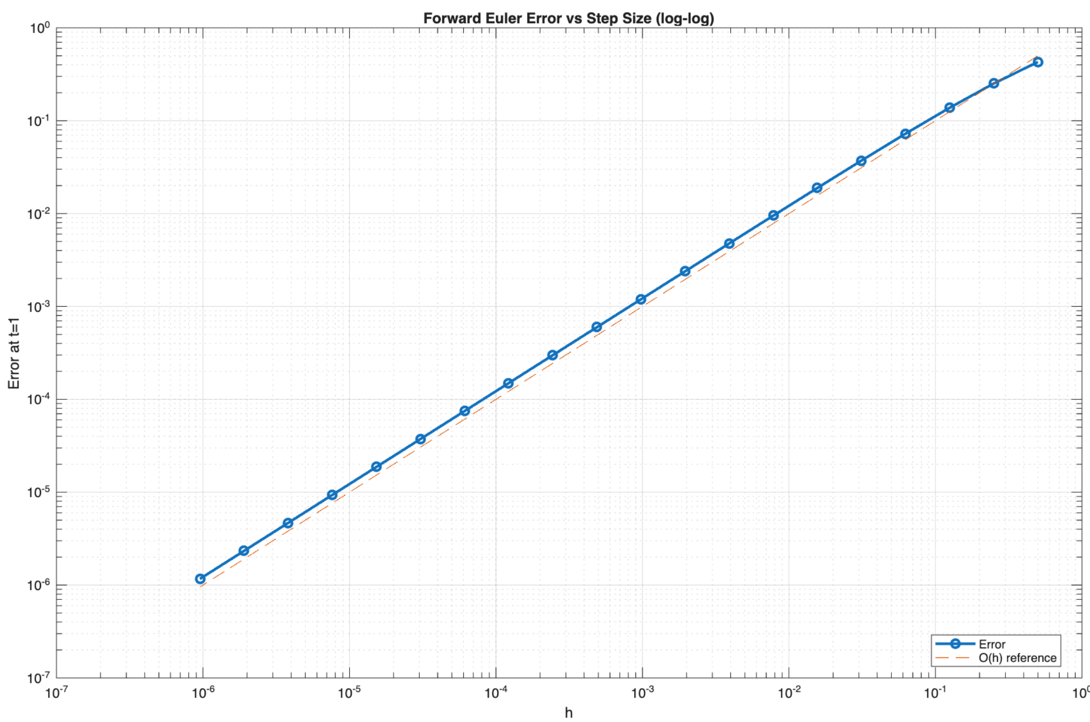

FIGURE 2. For $y' = \sin(t) + y$, a log-log plot of the Forward Euler's method approach based on step size.

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5.d. For 5.d. we do (b) and (c) again, but this time for the backwards Euler's method. The code follows below as

```
clear; clc;

yexact = @(t) 0.5*(exp(t) - sin(t) - cos(t));

a = 0;
b = 1;
```

```
Nh = 10;
h = (b-a)/Nh;

t = zeros(Nh+1,1);
y = zeros(Nh+1,1);

% initial condition
t(1) = 0;
y(1) = 0;

% Backward Euler method
for n = 1:Nh
    t(n+1) = t(n) + h;
    y(n+1) = ( y(n) + h*sin(t(n+1)) ) / (1-h);
end

y_exact_at_1 = yexact(1);
error_at_1 = abs(y(end) - y_exact_at_1);

fprintf('Backward Euler approximation at t=1: %.10f\n', y(end));
fprintf('Exact value at t=1: %.10f\n', y_exact_at_1);
fprintf('Absolute error at t=1: %.10f\n', error_at_1);
```

which yields the following output

```
Backward Euler approximation at t=1: 0.8026112881
Exact value at t=1: 0.6682542689
Absolute error at t=1: 0.1343570192
```

And our figure for the convergence of the backwards Euler's method follows as Figure **??**

5.e. When considering which to use, backwards or forwards Euler methods we have to think about the problem we are faced with. Generally, the forward Euler's method requires less effort as we get y values directly as a product of doing the iteration, but that trades off for stability of the solution. The backwards Euler's method requires algebraic manipulation from step to step, whereas the forwards Euler's method only requires calculation. Again the main trade-off is the stability of the solution.

Both forward Euler and backward Euler are first-order methods, so their errors decrease proportionally to h. On a log-log plot, this appears as an approximately straight line of slope 1. Forward Euler is simpler and faster since it is explicit and does not require solving an equation at each step. Backward Euler is implicit and therefore usually more expensive computationally, but it has better stability properties. For this problem, backward Euler can be solved algebraically at each step, so the extra cost is small, though still somewhat greater than forward Euler. In general, forward Euler is advantageous for simple nonstiff problems, while backward Euler is preferred when stability is more important, especially for stiff equations.

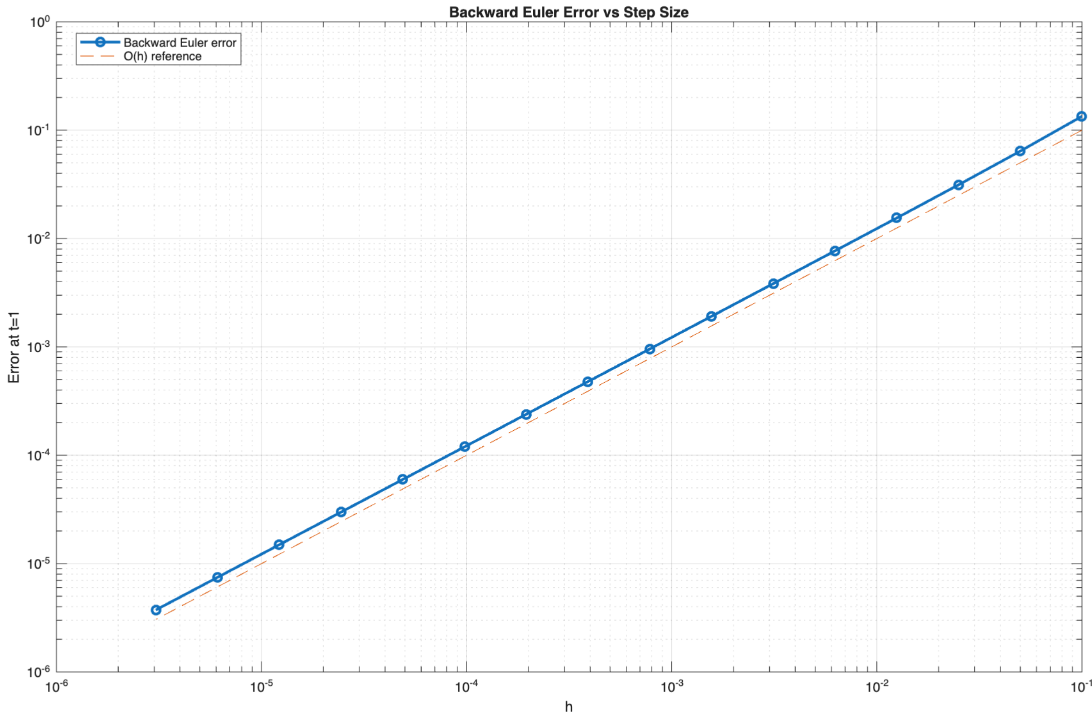

FIGURE 3. For $y' = \sin(t) + y$, a log-log plot of the Backwards Euler's method approach based on step size.

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